

MediTrack AI

Complete User Manual & System Documentation

Patient

Doctor

Admin



MediTrack AI is a professional healthcare ecosystem that integrates Artificial Intelligence to help patients analyze symptoms, manage medications, and connect seamlessly with verified healthcare professionals.

Firebase Auth	AI Inference Engine	iText7 PDF	Android 12-14
----------------------	----------------------------	-------------------	----------------------

Table of Contents

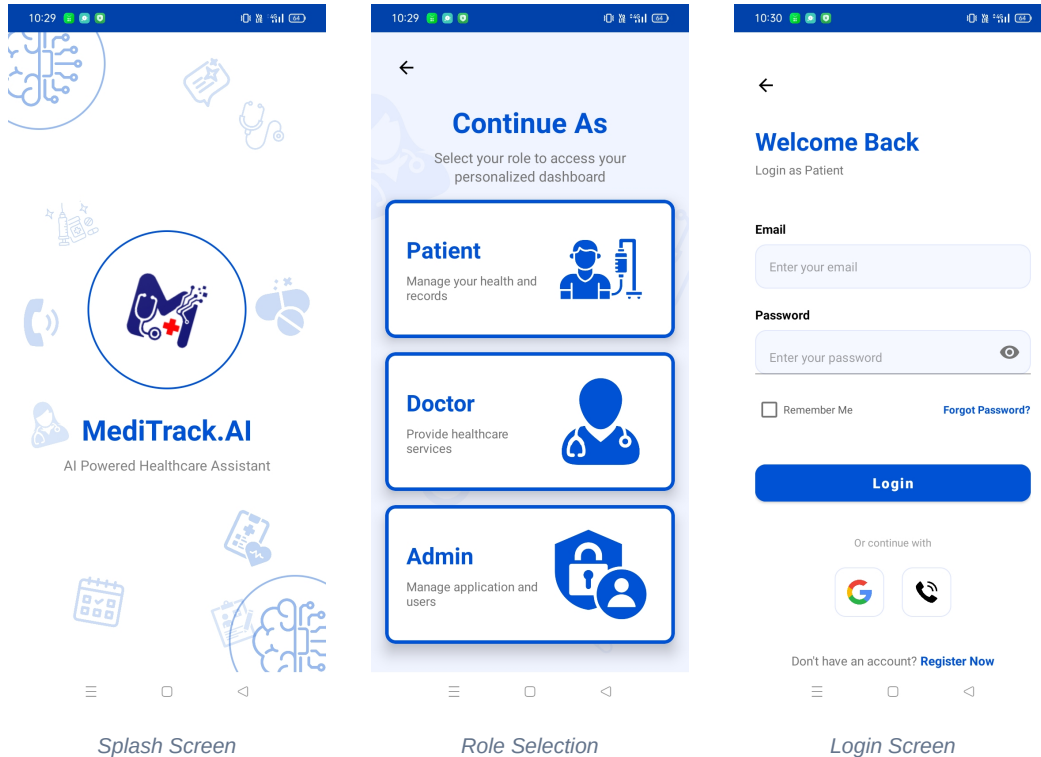
1.	Getting Started: Onboarding & Authentication
2.	Patient Journey: Managing Your Health
3.	Doctor Journey: Providing Care
4.	Admin Journey: System Oversight
5.	Core Technical Working
6.	App Flow Summary

1. Getting Started: Onboarding & Authentication

A. Splash Screen

Displays a high-quality rotating logo with a creative medical doodle background.

Logic: The system automatically detects if a user is already logged in. If yes, it redirects them to their specific dashboard; if not, it proceeds to Role Selection.



B. Role Selection

New users must choose one of three roles: **Patient**, **Doctor**, or **Admin**. This choice defines the entire app experience and available features.

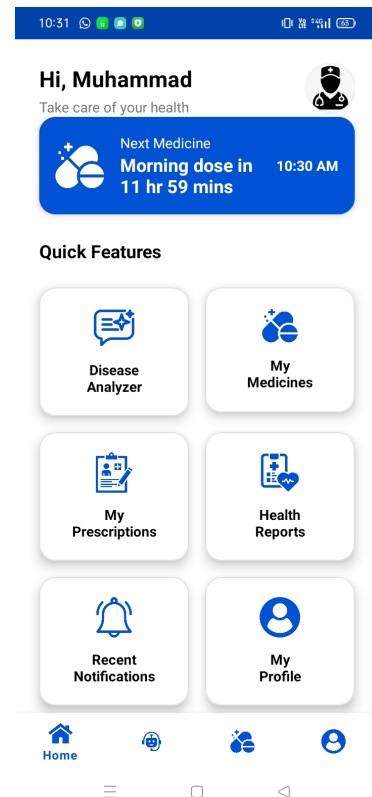
C. Registration & Login

Role	Required Fields
Patient	Name, Email, Phone Number
Doctor	Name, Email, Phone + Qualifications, Specialization, Years of Experience
Admin	Secured system-level credentials

2. Patient Journey: Managing Your Health

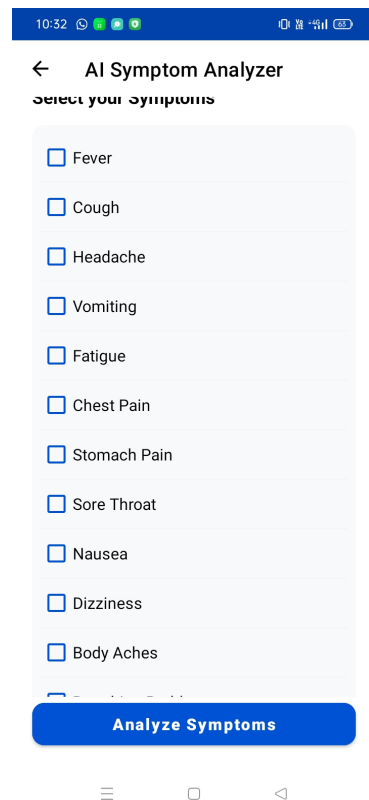
1. Patient Dashboard (Central Hub)

The Patient Dashboard serves as the central control center. It displays a real-time medicine countdown timer at the top, showing time remaining for the next scheduled dose. Quick access buttons provide navigation to AI Analyzer, Reports, and Prescriptions.

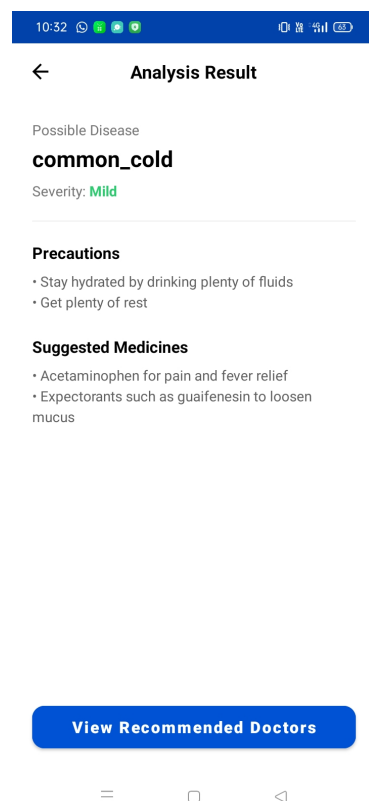


2. AI Disease Analyzer

Patients select symptoms (e.g., Fever, Cough, Chest Pain) from an interactive list. The AI Inference Engine processes these inputs and returns a Predicted Disease, Severity Level, and Necessary Precautions — all within seconds.



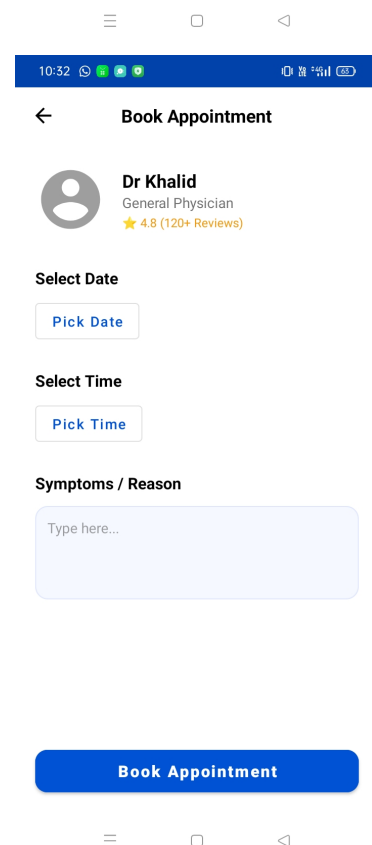
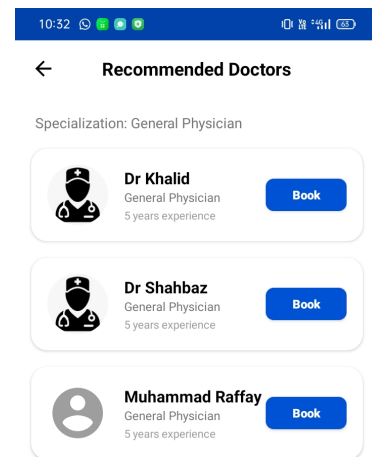
The image shows a mobile app interface for an "AI Symptom Analyzer". At the top, there's a status bar with the time 10:32 and various icons. Below that, a blue header bar contains a back arrow and the title "AI Symptom Analyzer". The main heading is "Select your symptoms". A list of symptoms is displayed, each with a blue checkbox: Fever, Cough, Headache, Vomiting, Fatigue, Chest Pain, Stomach Pain, Sore Throat, Nausea, Dizziness, and Body Aches. At the bottom of the list is a blue button labeled "Analyze Symptoms". The bottom of the screen shows the standard Android navigation bar with three icons: a hamburger menu, a home button, and a back arrow.



The image shows the "Analysis Result" screen of the mobile app. It has a blue header bar with a back arrow and the title "Analysis Result". The main content area is light gray. It starts with "Possible Disease" followed by "common_cold" in bold. Below that, "Severity: Mild" is shown in green. A horizontal line separates this from the "Precautions" section, which lists: "Stay hydrated by drinking plenty of fluids" and "Get plenty of rest". Another horizontal line separates this from the "Suggested Medicines" section, which lists: "Acetaminophen for pain and fever relief" and "Expectorants such as guaifenesin to loosen mucus". At the bottom is a blue button labeled "View Recommended Doctors". The bottom of the screen shows the standard Android navigation bar with three icons: a hamburger menu, a home button, and a back arrow.

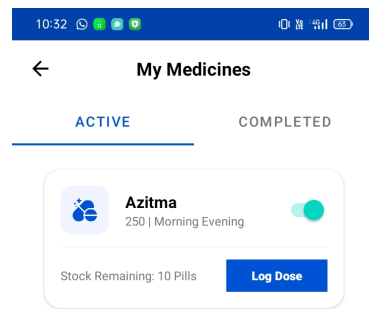
3. Booking Appointments

Based on the AI analysis results, patients can browse available doctors filtered by specialization and send appointment requests along with their symptom summary for the doctor's reference.

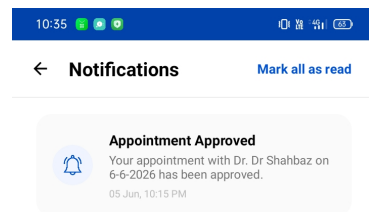


4. Medicine Management & Smart Reminders

Patients log their daily medications with dosage schedules. The app uses Android's AlarmManager to trigger loud notifications and alarms when it's time to take a pill — even when the app is closed or the phone is in sleep mode.



Add Medicine Dose



5. Digital Prescriptions

Patients can view all prescriptions issued by their doctors and download them as professional PDF files using the iText7 engine, enabling offline access to their complete medical records.

The screenshot shows the 'Create Prescription' screen. At the top, there's a status bar with the time 10:34 and various icons. Below the title bar, there's a back arrow and the title 'Create Prescription'. The patient's name 'Patient: sajid' is displayed. Under the 'Diagnosis' section, there's a text input field containing 'Flu / Influenza'. The 'Add Medicines' section has a plus icon. The 'Notes' section contains a text input field with the text 'Drink plenty of fluids...'. At the bottom, there's a large blue button labeled 'Save & Generate PDF'. The bottom navigation bar shows three icons: a menu, a home icon, and a back arrow.

AI Chatbot Assistant

An AI-powered chatbot is available to answer patient health queries in natural language, provide general medical information, and guide users through the app features for a seamless experience.

The screenshot shows the AI Chatbot Assistant interface. At the top, there's a status bar with the time 10:32 and various icons. Below the title bar, there's a back arrow, another back arrow, and a chat bubble icon with the text 'AI Chatbot'. A blue circular button with the text 'Hi' is visible. Below this, there's a message from the AI Assistant: 'Welcome to MediTrack AI. I'm here to help you with any medical-related questions or concerns you may have. Are you experiencing a specific health issue, looking for general wellness advice, or seeking information about a certain medical topic?'. The bottom of the screen shows a text input field with the placeholder 'Type a message...' and a blue button with a right arrow. The bottom navigation bar shows three icons: a menu, a home icon, and a back arrow.

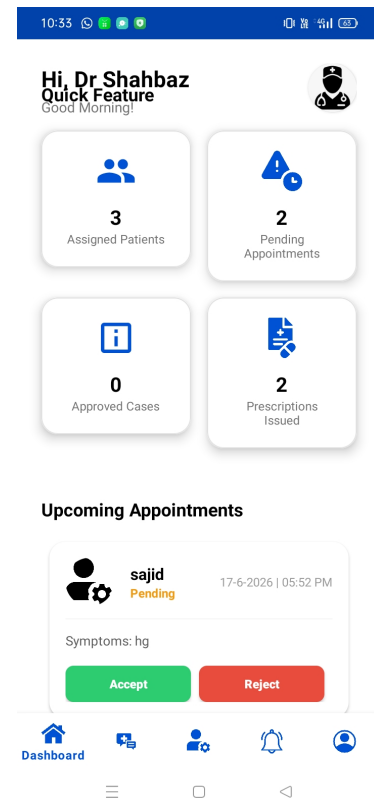
3. Doctor Journey: Providing Care

1. Verification Phase

After registration, a doctor's account enters a 'Pending' state. The doctor cannot access the dashboard until the Admin reviews and verifies their qualifications and credentials, ensuring platform safety.

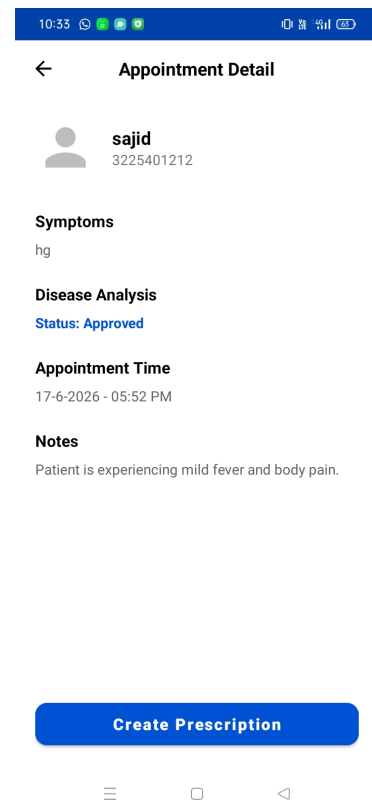
2. Doctor Dashboard

Provides a bird's-eye overview showing Total Patients, Today's Schedule, and Pending Appointment Requests in real time — helping doctors manage their workload efficiently.



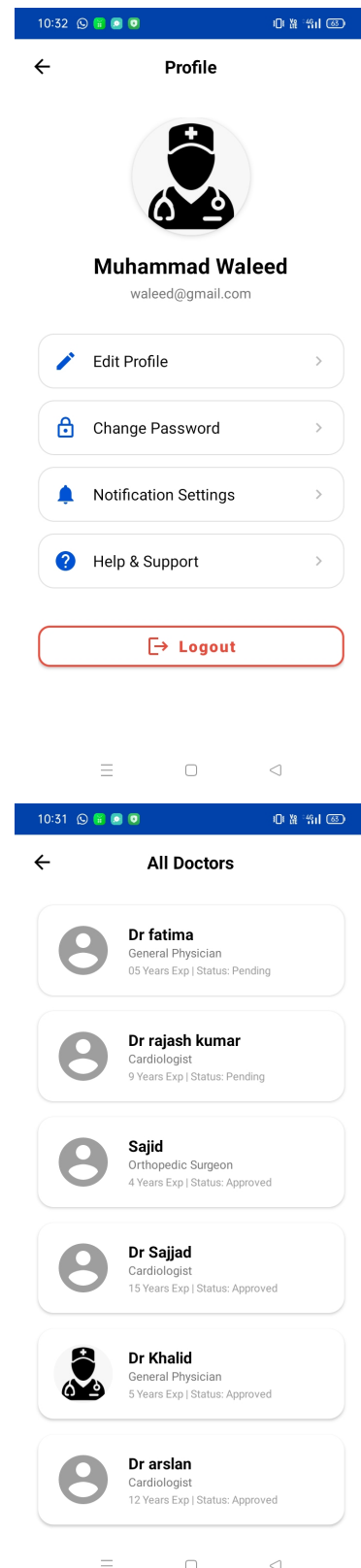
3. Appointment Management

Doctors receive real-time push notifications for new appointment requests. They can review patient symptoms and Accept or Reject requests based on their availability and specialization.



4. Doctor & Patient Profiles

Comprehensive profiles display doctor qualifications, specialization, experience, and contact information. Patients' profiles show their medical history and current condition for better-informed consultations.



5. Issuing Digital Prescriptions

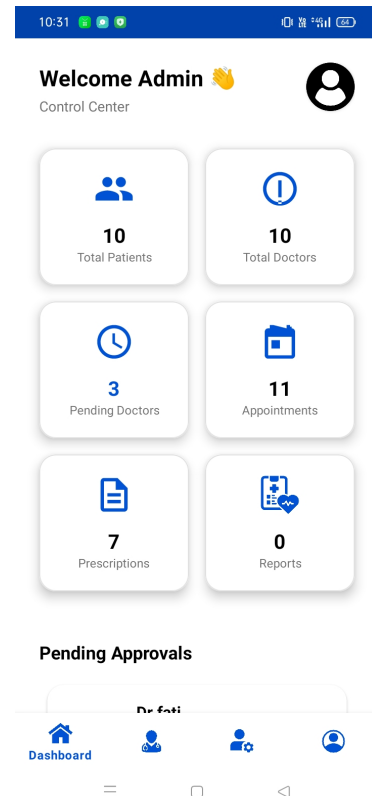
Once an appointment is approved, doctors use a built-in prescription form to prescribe medicines, set dosage schedules (Morning/Evening), and add clinical notes. Once submitted, the prescription instantly appears on the Patient's dashboard via Firebase real-time sync.

4. Admin Journey: System Oversight

The Admin is the **Superuser** who ensures the platform remains safe, verified, and well-organized for all users.

1. Admin Dashboard

Displays global platform statistics including: Total Registered Doctors, Total Active Patients, and System-Wide Prescriptions Issued — giving a complete view of the platform's health.



2. Doctor Approval Center (Critical Feature)

10:30







Admin Login

Login to Admin Portal

Email

Password

Login

The Admin has authority to monitor all user activities, manage accounts, and maintain overall platform integrity — ensuring MediTrack AI remains a trusted and safe healthcare environment.

5. Core Technical Working (Behind the Scenes)

AI Inference Engine

A logic-based AI model that maps symptom clusters to medical conditions with high accuracy. Trained on medical datasets to provide Predicted Disease, Severity Level, and Precautions.

Firebase Real-time Infrastructure

Ensures instant data synchronization across all users. When a doctor writes a prescription, the patient sees it in less than a second — no refresh needed.

Android Background Services

Optimized background services handle medicine reminders without draining the battery. Fully compatible with Android 12, 13, and 14, using AlarmManager for precise scheduling.

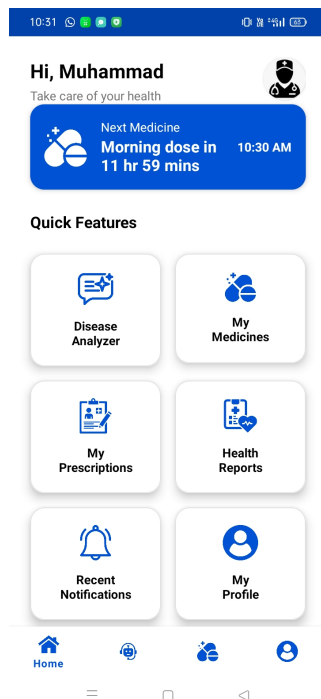
iText7 PDF Engine

Generates high-resolution, print-ready PDF documents for medical prescriptions and records, allowing patients to download and share their medical history offline.

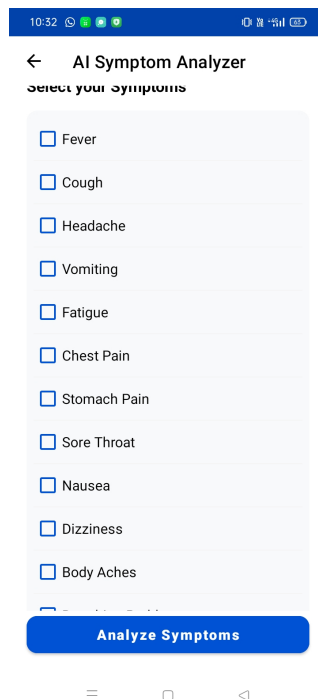
6. App Flow Summary

1	Onboarding	Select Role → Register → Login
2	Diagnosis	Patient uses AI Analyzer to understand symptoms
3	Consultation	Patient books a Doctor → Doctor Approves request
4	Treatment	Doctor issues Prescription → Patient receives Reminders & PDF
5	Security	Admin verifies all Doctors to ensure patient safety

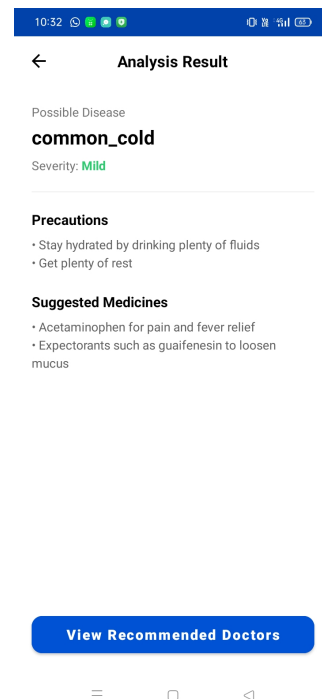
App Screenshots Gallery



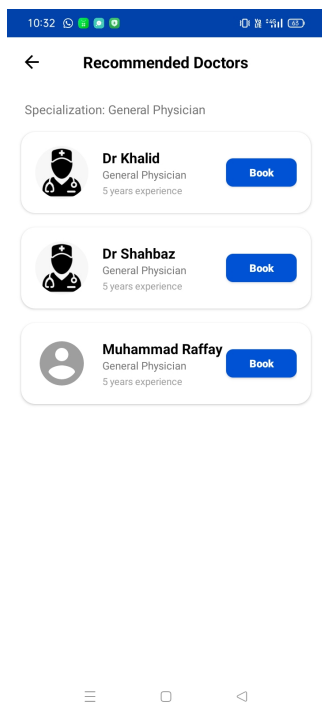
Patient Dashboard



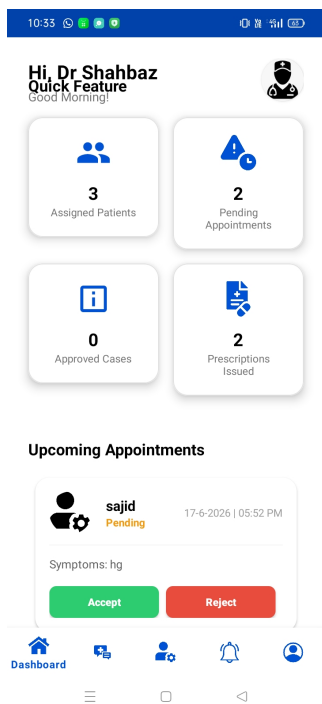
AI Symptom Analyzer



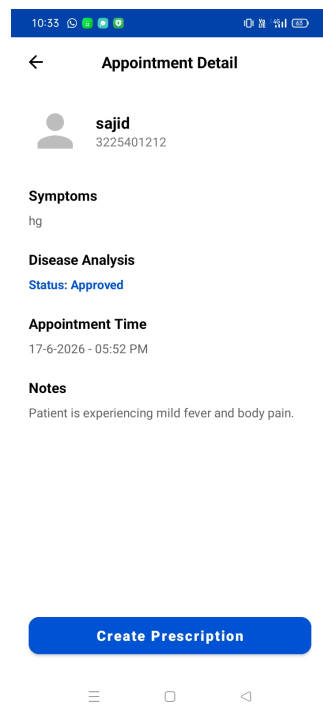
Analysis Result



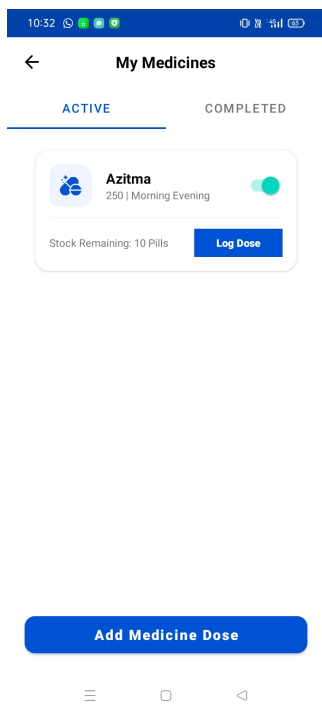
Recommended Doctors



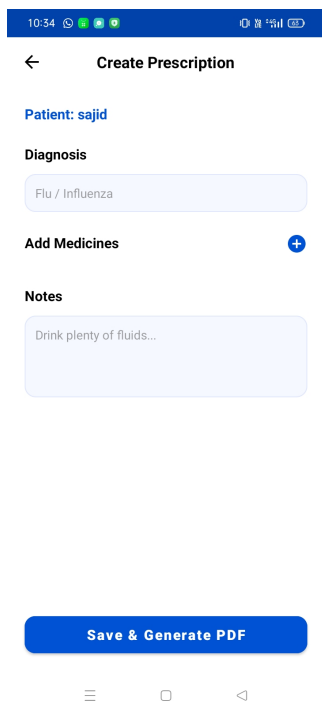
Doctor Dashboard



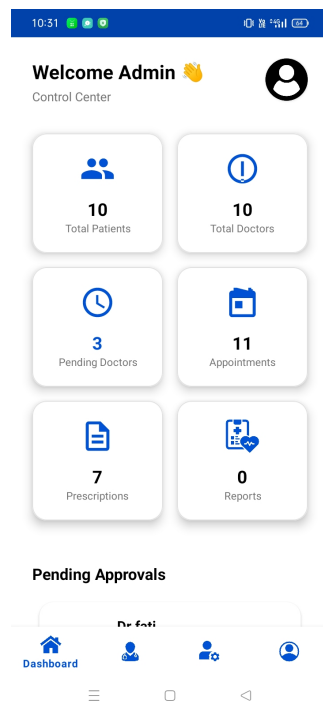
Appointment Details



Medicine Management



Prescription View



Admin Dashboard